Protocols and interpretation

Historical accuracy versus spectral diagnostics

Historical exact-loss evaluation uses 32 matched 2022 starts and 40 six-hour leads. Spectral evaluation uses eight previously inspected 2023 starts, 30–60° latitude in both hemispheres, and zonal wavenumbers 1–180. Neither is the pending year-round 2023 evaluation.

Objective and measurements

Training AMSE interpolates spherical spectral MSE (lambda=0) and full AMSE (lambda=1), using a global T179 transform. Verification uses zonal Fourier spectra. Exact GraphCast loss retains its original normalization, latitude, pressure and variable weights; losses are pooled before the reduction ratio.

October joint qualification

On the same eight spectral starts: exact GC loss ratio <1, mean physical-RMSE ratio <1, and mean relative-power-error ratio <1; additionally no selected primitive field/day may exceed 2% RMSE harm versus GC. RMSE/power summaries use nine fields and days 1, 3, 7, 10. Power error is averaged over spectral bins then field/day cells.

Baseline and uncertainty limits

Historical comparisons flag numerical baseline drift and use each model’s paired GC baseline. October scores use their phase’s canonical GC reference; controls vary between re-evaluations. Do not average repeated baselines or interpret tiny differences as established gains. Development selection has no confirmatory uncertainty claim.

Power is not accuracy

Matching power does not ensure the correct location or phase of weather features. Coherence and spectral-error decomposition in the appendix expose this distinction. No new forecast inference or training was run to prepare this report.